

E-Commerce Backend Performance & Revenue Analytics

Using PostgreSQL and Power BI

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Tools Used: PostgreSQL (pgAdmin), Power BI Desktop

Project Type: Business Intelligence & Data Analytics

1. Project Overview

The objective of this project is to analyze e-commerce operational and sales data to generate business insights and support data-driven decision-making. PostgreSQL was used for data extraction and analysis, while Power BI was used to create interactive dashboards for business users.

2. Business Objectives

The project aims to:

- Analyze overall sales performance.
- Identify top-performing products and categories.
- Evaluate customer purchasing behavior.
- Monitor inventory levels and low-stock products.
- Track monthly and daily sales trends.
- Analyze coupon usage and its impact.
- Measure product performance using sales and ratings.
- Evaluate shipping performance.
- Generate customer insights and geographic distribution.
- Build interactive Power BI dashboards.
- Implement Row-Level Security (RLS).

3. Database Tables Used

Table Name	Description
customers	Customer information
orders	Order details
order_items	Products sold in each order
products	Product information
categories	Product categories
reviews	Customer product ratings
shipping_data	Shipping information
coupons	Discount coupon information
inventory_log	Inventory movement records

4. SQL Analysis

```
SELECT current_database();  
SELECT table_name  
FROM information_schema.tables  
WHERE table_schema = 'public';
```

	table_name name
1	categories
2	products
3	customers
4	orders
5	inventory_logs
6	reviews
7	order_items
8	shipping_deta...
9	coupons

SELECT * FROM categories;

SELECT * FROM products;

SELECT * FROM customers;

SELECT * FROM orders;

SELECT * FROM inventory_logs;

SELECT * FROM reviews;

SELECT * FROM order_items;

SELECT * FROM shipping_details;

SELECT * FROM coupons;

To check nulls

SELECT COUNT(*) - COUNT(customer_id) FROM orders;

SELECT COUNT(*) - COUNT(total_amount) FROM orders;

SELECT COUNT(*) - COUNT(product_name) FROM products;

SELECT COUNT(*) - COUNT(email) FROM customers;

Total Revenue

```
SELECT SUM(total_amount) AS total_revenue  
FROM orders;
```

		total_revenue numeric	
1		17846538.04	

Total Orders

```
SELECT COUNT(order_id) AS total_orders  
FROM orders;
```

		total_orders bigint	
1		5610	

Avg Order Value

```
SELECT AVG(total_amount) AS avg_order_value  
FROM orders;
```

		avg_order_value numeric	
1		3181.2010766488413547	

Top Products by Revenue

```
SELECT
p.product_name,
SUM(oi.item_total) revenue
FROM order_items oi
JOIN products p
ON oi.product_id=p.product_id
GROUP BY p.product_name
ORDER BY revenue DESC;
```

	product_name character varying (255)	revenue numeric
1	Canon EOS R6 Mark II - Model A	1107639.20
2	Canon EOS R6 Mark II - Model C	677522.43
3	MacBook Pro M3 14-inch - Model A	556955.20
4	Dell XPS 13 Laptop - Model B	551066.66
5	iPhone 15 Pro Max 256GB - Model B	516572.33
6	Canon EOS R6 Mark II - Model B	514477.49
7	MacBook Pro M3 14-inch - Model C	420978.45
8	iPhone 17 Pro Max 512GB - Model B	414984.63
9	Samsung Galaxy S24 Ultra - Model C	396111.59
10	iPhone 17 Pro Max 512GB - Model A	391037.67

Top Categories

```
SELECT  
c.category_name,  
SUM(oi.item_total) revenue  
FROM order_items oi  
JOIN products p  
ON oi.product_id=p.product_id  
JOIN categories c  
ON p.category_id=c.category_id  
GROUP BY c.category_name  
ORDER BY revenue DESC;
```

	category_name character varying (100) 	revenue numeric 
1	Electronics	10202228.50
2	Home & Kitchen	2540697.78
3	Beauty	2388135.60
4	Apparel	1809203.97
5	Books	1349601.29

Customer Lifetime Value (CLTV)

```
SELECT  
customer_id,  
SUM(total_amount) AS lifetime_value  
FROM orders  
GROUP BY customer_id  
ORDER BY lifetime_value DESC;
```

	customer_id  integer	lifetime_value  numeric
1	676	87687.00
2	1057	75839.00
3	727	62193.22
4	458	60667.22
5	1061	57726.00
6	1012	56651.45
7	15	56229.00
8	416	52822.23
9	161	51071.30
10	1066	50903.90

Inventory Status

```
SELECT
product_name,
stock_quantity
FROM products
ORDER BY stock_quantity ASC;
```

	product_name character varying (255) 🔒	stock_quantity integer 🔒
1	The Ordinary Niacinamide 10% - Model C	5
2	AirPods Pro 2nd Gen - Model C	5
3	The Great Gatsby - Model A	6
4	iPad Air M2	9
5	Estée Lauder Advanced Night Repair - Mod...	11
6	La Mer Crème de la Mer - Model C	11
7	iPhone 15 Pro Max 256GB - Model B	12
8	La Mer Crème de la Mer - Model C	13
9	Levis 501 Original Jeans - Model C	13
10	Principles by Ray Dalio - Model B	16

Repeat Customers

```
SELECT  
customer_id,  
COUNT(order_id) AS orders_count  
FROM orders  
GROUP BY customer_id  
HAVING COUNT(order_id) > 1;
```

	customer_id integer	orders_count bigint
1	652	5
2	273	5
3	51	3
4	1091	7
5	951	2
6	70	3
7	839	4
8	1075	3
9	350	5
10	758	5

Coupon Performance

```
SELECT
coupon_id,
COUNT(*) AS usage_count,
SUM(total_amount) AS revenue
FROM orders
WHERE coupon_id IS NOT NULL
GROUP BY coupon_id;
```

	coupon_id 	usage_count 	revenue 
	integer	bigint	numeric
1	6	121	292543.10
2	1	135	414617.60
3	3	130	262216.00
4	5	127	286952.60
5	2	125	290188.60
6	4	130	181415.00

AVG Product Rating

```
SELECT
p.product_name,
AVG(r.rating) AS avg_rating
FROM products p
JOIN reviews r
ON p.product_id = r.product_id
GROUP BY p.product_name;
```

	product_name character varying (255)	avg_rating numeric
1	Tefal Non-Stick Set 10pc - Model B	4.6666666666666667
2	The Psychology of Money - Model A	4.5000000000000000
3	Fenty Beauty Gloss Bomb - Model C	5.0000000000000000
4	Nike Air Jordan 1 Low - Model A	4.4375000000000000
5	Dell XPS 13 Laptop - Model B	4.3333333333333333
6	MacBook Pro M3 14-inch - Model A	5.0000000000000000
7	iPhone 17 Pro Max 512GB - Model C	3.8333333333333333
8	Bialetti Moka Express - Model C	4.5000000000000000
9	Sapiens: A Brief History	3.0000000000000000
10	Instant Pot Duo 7-in-1 - Model B	4.3333333333333333

Shipping Performance

```
SELECT
shipping_method,
COUNT(*) AS total_deliveries
FROM shipping_details
GROUP BY shipping_method;
```

	shipping_method character varying (50) 	total_deliveries bigint 
1	Aramex	610
2	Aramex Express	5000

```
SELECT
shipping_city,
COUNT(*) AS deliveries
FROM shipping_details
GROUP BY shipping_city
ORDER BY deliveries DESC;
```

	shipping_city character varying (100) 	deliveries bigint 
1	Dubai	1530
2	Ras Al Khaimah	1075
3	Abu Dhabi	1065
4	Ajman	1046
5	Sharjah	894

For an Executive Summary / Overall Sales Performance analysis.

```
SELECT
    COUNT(order_id) AS total_orders,
    SUM(total_amount) AS total_revenue,
    AVG(total_amount) AS average_order_value,
    MIN(order_date) AS first_order_date,
    MAX(order_date) AS last_order_date
FROM orders;
```

	total_orders bigint	total_revenue numeric	average_order_value numeric	first_order_date timestamp without time zone	last_order_date timestamp without time zone
1	5610	17846538.04	3181.2010766488413547	2026-01-01 00:14:57	2026-06-15 00:00:00

TO Combines data from multiple tables into one dataset.

```
SELECT
    o.order_id,
    o.order_date,
    o.total_amount,
    o.status,
    c.customer_id,
    c.first_name,
    c.last_name,
    c.city,
    p.product_id,
    p.product_name,
```

```
cat.category_name,  
oi.quantity,  
oi.item_total  
FROM orders o  
JOIN customers c ON o.customer_id = c.customer_id  
JOIN order_items oi ON o.order_id = oi.order_id  
JOIN products p ON oi.product_id = p.product_id  
JOIN categories cat ON p.category_id = cat.category_id;
```

5. Power BI Dashboard Development

Data Modeling

Relationships created:

- Customers → Orders
- Orders → Order Items
- Products → Order Items
- Categories → Products
- Orders → Shipping Data
- Products → Reviews
- Calendar → Orders

6. DAX Measures

- Total Revenue
- Total Orders
- Total Quantity Sold
- Average Rating
- Average Order Value
- Repeat Customers
- MOM Growth%
- Total Customers

- Customer Lifetime Value(CLTV)
- Total Products
- Total Inventory
- Total Shipments
- Average Shipping Cost
- Coupon Usage Count
- Low Stock Products

6. Dashboard Pages

Executive Summary

KPIs:

- Total Revenue
- Total Orders
- Total Quantity Sold
- Average Rating
- Average Order Value
- Repeat Customers
- MOM Growth%
- Total Customers
- Customer Lifetime Value(CLTV)
- Total Products
- Total Inventory
- Total Shipments
- Average Shipping Cost
- Coupon Usage Count
- Low Stock Products

Visuals:

- **Total Revenue by Month**
- **Total Revenue by Category**
- **Total Revenue by Top 10 Product**
- **Total Orders by category**
- **Total Customers by City**
- **Customer Acquisition Trend**
- **Total Revenue by Top 5 Customers**

- **Top 10 Product Quantity Sold**
- **Top 10 Product Rating**
- **Stock Availability**
- **Inventory Distribution**
- **Inventory Movements**
- **Coupon Usage**
- **Shipment By City**
- **Shipping Method Analysis**
- **Total Shipping Cost By Method**

7. Row-Level Security (RLS)

Implemented RLS using city-based filtering.

Example:

Public customers[city] ='Dubai' and Named it Dubai Manager

This ensures users can only view customer and sales data related to their assigned city.

8. Challenges Faced

1.Database access was read-only.

2.SQL Views could not be created due to permission restrictions (permission denied for schema public).

3.Equivalent business logic was implemented using SQL queries and Power BI data modeling.

9.Business Insight

- Electronics dominate total revenue (10.2 M of 18.29 M total), showing heavy dependence on one category.
- Customer acquisition peaked in April (260) but dropped sharply in June (117). Repeat customers (1 K) form a solid base, yet growth momentum is slowing.
- Aramex Express handles 83 % of shipments but costs 5x more than Aramex.

- Customer and shipment maps show strong clusters in Dubai and Abu Dhabi, with weaker engagement in northern emirates.
- The dashboards show a large number of customers and shipments, but the average order value (AOV) is relatively low compared to total revenue.

10.Recommendations

- Sales are mostly coming from Electronics. Books and Apparel are weaker. The company should promote these categories more and adjust prices if needed, so revenue is balanced and not dependent on one category
- Customer growth is slowing in some months. The company should run special campaigns during these low-growth periods, and study why customers leave. This will help keep monthly growth above 35%.
- Aramex Express is handling most shipments but at a much higher cost. The company should negotiate better rates or move some shipments to cheaper carriers. This will lower delivery costs and improve profit margins.
- Expand marketing and delivery coverage in Sharjah, Ajman, and Ras Al Khaimah to capture untapped demand and balance regional sales.
- Customers often place small orders. The company should encourage bigger purchases by offering free shipping limits, product bundles, or premium options. This way, revenue will grow without needing more shipments

Conclusion

The dashboards give a clear picture of sales, customers, inventory, and shipping. They show where the business is strong and where improvements are needed. By promoting weaker categories, keeping customers loyal, managing stock better, and reducing shipping costs, the company can grow steadily and improve profits.
